

Eliyas Abraha

✉ eliyaskidane@gmail.com ☎ +31634451818 📍 Enshede, Netherlands
🔗 <https://eliyaskidnae.github.io/> 🔗 [linkedin.com/in/eliyas-abraha/](https://www.linkedin.com/in/eliyas-abraha/) 🐙 github.com/eliyaskidnae

👤 PROFILE

Robotics researcher with experience in 3D perception, sensor fusion, SLAM, navigation, and real-time autonomous systems. Interested in developing reliable robotic systems that combine robust perception, adaptive planning, and uncertainty-aware decision-making for real-world field and safety-critical applications.

📁 WORK EXPERIENCE

02/2025 – Present Enshede, Netherlands	Saxion University of Applied Sciences Smart Mechatronics and Robotics (SMART)  Robotics Research Engineer - Developed perception and autonomy pipelines for robotic systems using camera, LiDAR, and IMU data, supporting environment understanding, navigation, and safe operation in real-world robotic applications. - Applied 3D computer vision, sensor fusion, and mathematical modeling to support SLAM, tracking, registration, calibration, 3D reconstruction, and scene understanding for autonomous systems. - Implemented estimation and optimization techniques, including Extended Kalman Filtering and factor-graph-based optimization, to improve state estimation and increase robustness under uncertain sensor observations. - Contributed to robotics experiments through sensor setup, calibration, data collection, field testing, evaluation, and analysis of perception and navigation system performance.
07/2024 – 08/2025 Böhmenkirch, Germany.	Farming Revolution GmbH  Robotics Research Engineer - Worked on agricultural autonomous navigation stacks, including robot localization, global and local path planning, behavior planning, and navigation decision-making for field robotics applications. - Worked on perception and data preparation tasks for autonomous weeding robots, including image/data annotation support for crop, weed, and plant detection workflows. - Designed a row-based coverage path planning algorithm tailored to field geometry, ensuring efficient and uniform traversal for precision weeding.
02/2020 – 2023 Mekelle, Ethiopia	Lion International Bank  Software Engineer - Ensured reliability of database-backed systems by debugging and optimizing backend services, data pipelines, and storage layers. - Delivered IT support and digital infrastructure training to staff, increasing operational reliability and client satisfaction.

🎓 EDUCATION AND TRAINING

09/2023 – 06/2025 Spain	Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFROS) Erasmus Mundus Joint Master in Intelligent Field Robotic Systems (IFROS)  Studied Autonomous Systems, Probabilistic Robotics, Machine Learning, and Robot Manipulation.
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- Specialized in Computer Vision ,Deep Neural Network Development and 3D Sensing and Sensor Fusion.
- GPA 9.1/10

09/2014 – 07/2019
Ethiopia

B.Sc. in Computer Science and Engineering

Mekelle University

- Algorithm Design, Data structure, Linear Algebra, AI, and Computer Network..
- GPA 3.92/4



TECHNICAL SKILLS

Programming Languages

Python, C++, MATLAB, JavaScript
(Node.js, Vue.js)

Databases Management

MongoDB and MySQL

Development Tools

Docker, Git, Linux, VS Code,
Conda

Robotics Framework

ROS1/ROS2, Gazebo,
Rviz,GTSAM,Behavioral Trees

Computer Vision and AI

OpenCV, PCL, PyTorch, YOLO

Sensors and Perception

LiDAR, GPS, Camera (RGB/RGB-
D), IMU, Sensor Fusion

Geospatial Libraries

Shapely, GeoPandas,QGIS

Document writing

LATEX and Microsoft Office
tools



HONOURS AND AWARDS

01/05/2019

Outstanding Performance in Huawei ICT Competition Global Final

Huawei Academy



PUBLICATIONS

LiDAR-Inertial Localization with Prior 3D Maps in GNSS-Challenged Environments [↗](#)

Research paper based on Master's thesis — First draft completed; currently under revision for potential submission.



PROJECTS

09/2025 – 02/2026

Autonomous Volume Estimation of Bulk Feed in Unstructured Environments [↗](#)

Developed a computer vision pipeline for estimating the volume of bulk cow feed in unstructured stable environments. The project focused on detecting and segmenting feed regions from complex backgrounds such as floors, walls, and possible occlusions. RGB and depth-based data were processed to extract 3D information, separate the feed pile from the surrounding environment, and estimate feed geometry for volume calculation.

02/2025 – 06/2025

Robust-LiDAR-Inertial-Re-Localization(ROS2 and C++) [↗](#)

This thesis presents a robust and real-time localization framework for GNSS-denied environments by fusing LiDAR-Inertial Odometry (FAST-LIO2) with multithreaded NDT-based map matching using a sliding-window factor graph.

02/2024 – 05/2024

Pose Based SLAM using ICP LIDAR Scan Matching [↗](#)

Autonomous Exploration of Unknown Environments Using Dubins Path-Oriented RRT* [↗](#)

Stereo Vision and Object Depth Estimation [↗](#)

This project presents an economical alternative to LiDAR sensors for per-pixel depth estimation in drone applications. By utilizing block matching and Semi-Global Block Matching (SGBM) algorithms, we showcase how stereo computer vision techniques can accurately determine depth information.

Monocular Visual Odometry with KITTI Dataset 

Custom Object Detection using CNN and Image Captioning 

09/2025 – 01/2026

Practical Guide to SLAM for Industrial Applications 

Prepared and structured a practical course on SLAM focused on real-world industrial use cases, including benchmarking and sensor calibration.

REFERENCES

Mohammad ALDIBAJA, Associate Professor, Leader of Sensing and Perception Research Lines, at Saxion University
m.adlibaja@rug.nl, +31 6 41832724

Dr. ir. W.J. (Wilco) Bonestroo, Associate Professor, Cognition & AI, for Smart Mechatronics and Robotics Saxion University of Applied Sciences at saxion university
w.j.bonestroo@saxion.nl, +31 6 1886 7199

Prof Narcis Palomeras, Professor, University of Girona
narcis.palomeras@udg.edu, +34 683578554

Maurice Gohlke, CEO and Founder, farming revolution GmbH
maurice.gohlke@farming-revolution.com, +49 176 57752789